

North American Actuarial Journal

ISSN: 1092-0277 (Print) 2325-0453 (Online) Journal homepage: www.tandfonline.com/journals/uaaj20

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To cite this article: Yichun Chi, X. Sheldon Lin & Ken Seng Tan (2017) Optimal Reinsurance Under the Risk-Adjusted Value of an Insurer's Liability and an Economic Reinsurance Premium Principle, North American Actuarial Journal, 21:3, 417-432, DOI: 10.1080/10920277.2017.1302346

To link to this article: https://doi.org/10.1080/10920277.2017.1302346

Published online: 04 Aug 2017.

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North American Actuarial Journal, 21(3), 417–432, 2017
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ISSN: 1092-0277 print / 2325-0453 online
DOI: 10.1080/10920277.2017.1302346

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Optimal Reinsurance Under the Risk-Adjusted Value of an Insurer's Liability and an Economic Reinsurance Premium Principle

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In this article, an optimal reinsurance problem is formulated from the perspective of an insurer, with the objective of minimizing the risk-adjusted value of its liability where the valuation is carried out by a cost-of-capital approach and the capital at risk is calculated by either the value-at-risk (VaR) or conditional value-at-risk (CVaR). In our reinsurance arrangement, we also assume that both insurer and reinsurer are obligated to pay more for a larger realization of loss as a way of reducing ex post moral hazard. A key contribution of this article is to expand the research on optimal reinsurance by deriving explicit optimal reinsurance solutions under an economic premium principle. It is a rather general class of premium principles that includes many weighted premium principles as special cases. The advantage of adopting such a premium principle is that the resulting reinsurance premium depends not only on the risk ceded but also on a market economic factor that reflects the market environment or the risk the reinsurer is facing. This feature appears to be more consistent with the reinsurance market. We show that the optimal reinsurance policies are piecewise linear under both VaR and CVaR risk measures. While the structures of optimal reinsurance solutions are the same for both risk measures, we also formally show that there are some significant differences, particularly on the managing tail risk. Because of the integration of the market factor (via the reinsurance pricing) into the optimal reinsurance model, some new insights on the optimal reinsurance design could be gleaned, which would otherwise be impossible for many of the existing models. For example, the market factor has a nontrivial effect on the optimal reinsurance, which is greatly influenced by the changes of the joint distribution of the market factor and the loss. Finally, under an additional assumption that the market factor and the loss have a copula with quadratic sections, we demonstrate that the optimal reinsurance policies admit relatively simple forms to foster the applicability of our theoretical results, and a numerical example is presented to further highlight our results.

1. INTRODUCTION

Reinsurance is one of the key risk management tools for insurance companies to manage its risk. Reinsurance could strategically reduce an insurance company's risk exposure, enable it to underwrite more businesses, and stabilize earnings over time, just to name some of the potential benefits. As such, reinsurance should be integrated into the insurance company's prudent risk management strategy and exploited fully and optimally. Since the seminal work of Borch (1960), the quest for optimal contracts has attracted significant attention from both academics and practitioners, and a variety of optimal reinsurance policies have been devised by either minimizing some risk measure of an insurer's risk exposure or maximizing the expected utility of the final wealth of a risk-averse insurer. See, for example, Borch (1960), Arrow (1963), Raviv (1979), Huberman et al. (1983), Young (1999), Kaluszka (2001), Kaluszka and Okolewski (2008), Bernard and Tian (2009), and references therein. In particular, optimal reinsurance models based on value-at-risk (VaR) and conditional value-at-risk (CVaR) have been extensively studied recently in part due to their prevalent use in finance and insurance. See Cai and Tan (2007), Cai et al. (2008), Balbás et al. (2009), Tan et al. (2009), Cheung (2010), Chi and Tan (2011, 2013), and Chi and Lin (2014), just to name a few.

In insurance economics, the expected value principle is the most commonly adopted assumption for calculating the reinsurance premium. Its popularity stems from its simplicity and its tractability; optimal solutions usually can be derived explicitly for many

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reinsurance models. See, for example, Borch (1960), Arrow (1963), and Van Heerwaarden et al. (1989). As observed by Venter (1991), a potential drawback of using such a simple premium principle lies in its inconsistency with market practice, particularly with respect to property and casualty reinsurance pricing. As a result, many other premium principles have been introduced and analyzed. Young (2004), for example, catalogs as many as 11 of these premium principles. The generalization of these sophisticated premium principles stimulates more innovations and creates more challenges in optimal reinsurance problems. For instance, optimal reinsurance is studied by Kaluszka (2001) for mean-variance premium principles and by Young (1999) for Wang's premium principle. More generally, Chi and Tan (2013) show that one-layer reinsurance is always optimal under VaR and CVaR risk measures if the reinsurance premium principle preserves the stop-loss order. Their results are further generalized by Chi (2012), who addresses the optimal reinsurance with reinsurance premium principles that are convex order preserving.

In the afore-mentioned studies, the reinsurance premium is calculated based only on the ceded loss. Bühlmann (1980), on the other hand, argues that the reinsurance market is incomplete, so the reinsurance premium should depend not only on the ceded loss itself but also on the existing market environment. For this reason, he advocates an economic pricing of reinsurance contracts that reflects the risk exposure not only attributed to the potential losses but also from the market itself. Motivated by the equilibrium pricing model of Bühlmann (1980), Wang (2002) introduces a premium pricing framework based on exponential tilting, which in turn has been extended to economic weighted pricing by Furman and Zitikis (2009) and the multivariate analog by Zhu et al. (2016). Under an economic premium principle, not only that the reinsurance premium is typically no longer law-invariant, but it also depends on the joint distribution of the ceded loss and the underlying economic factors reflecting the market environment. Another advantage of the economic premium principle is that it can be perceived as a general class of premium principles that encompasses many popular premium principles such as net, modified covariance, size-biased, Esscher, Aumann-Shalpey, Kamps, excess-of-loss, distorted, proportional hazard, modified tail covariance, and conditional tail expectation (CTE) premium principles as special cases. It is also worth noting that the cost-of-capital premium calculation principle introduced by Merz and Wüthrich (2014) is a special case of the economic premium principle. Despite its viability, to the best of our knowledge there are very limited studies using the economic premium principle in the context of optimal reinsurance problems. It is precisely the objective of this article to fill this gap.

In this article, we study the design of an optimal reinsurance policy from the perspective of an insurer, where the reinsurance premium is calculated by an economic premium principle. Mathematically, the reinsurance premium is set as the expected ceded loss weighted by a market factor. To reduce ex post moral hazard, we assume that both parties in a reinsurance arrangement are obligated to pay more for a larger realization of loss, as suggested by Huberman et al. (1983). The adopted objective of the optimal reinsurance model is to minimize the value of the insurer's risk-adjusted liability, where the valuation is carried out by a cost-of-capital approach proposed by Swiss Federal Office of Private Insurance (2006). When the capital at risk is calculated by the VaR or CVaR risk measure, we show that strategically it is optimal for the insurer to purchase a piecewise linear reinsurance policy, that is, the derivative of whose ceded loss function is an indicator function. We find that although the optimal reinsurance policies under both of these two risk measures are alike in some ways, they are also different in many other aspects, particularly in ceding tail risk. This finding is novel and important, especially under the current regime of transitioning from VaR to CVaR for quantifying the regulatory capital. In addition, we find that the optimal reinsurance heavily relies not only on the distribution function of the market factor, but also on the dependence between the loss and the market factor. As a result, the effect of the market factor on optimal reinsurance design is analyzed in detail by a comparative analysis. Finally, to illustrate the applicability of our theoretical results, explicit and relatively simple forms of the optimal contracts are derived under the assumption that the market factor and the loss have a bivariate copula with quadratic sections.

The optimal reinsurance model analyzed in this article is closely related to Chi (2012) and Asimit et al. (2013). There are, however, some important differences. First, although the formulations of the optimal reinsurance models are similar in that all of them have the same objective function, the adopted reinsurance premium principles are vastly different. They usually assume that the reinsurance premium principles are law-invariant and convex order preserving. In contrast, the economic premium principle that we are considering in this article usually does not satisfy these two properties. Second, although all the optimal reinsurance policies are piecewise linear, our optimal contracts are usually much more complicated than theirs. Third and most importantly, our optimal reinsurance policies depend heavily on a market factor via the economic premium principle. This, in turn, enables us to conduct a comparative analysis and provide important insights on the effects of the market factor and the dependence between the market factor and the loss on optimal reinsurance.

The rest of this article is organized as follows. Section 2 introduces the optimal reinsurance model with an economic premium principle that we will investigate. The model is solved and the optimal reinsurance is derived explicitly in Section 3. Section 4 carries out a comparative analysis of the optimal contract with respect to the market factor. To illustrate the applicability of our theoretical results, optimal reinsurance policies with simple forms are obtained in Section 5 under a special dependence structure between the market factor and the loss. Section 6 concludes the article and the Appendix collects the proofs to the theorems and propositions established in the article.

2. THE MODEL

Let X be the loss an insurer faces over a given time period. It is natural to assume that X is a non-negative random variable on the probability space $(\Omega, \mathscr{F}, \mathbb{P})$ with cumulative distribution function (c.d.f.) $F_X(x) \triangleq \mathbb{P}(X \leq x)$ and $\mathbb{E}[X] < \infty$. The main concern of reinsurance design is how to optimally split X into $f(X)$ and $R_f(X)$, where $f(X)$ represents the portion of loss that is ceded to a reinsurer while $R_f(X)$ is the residual loss retained by the insurer (cedent). Thus, $f(x)$ and $R_f(x)$ are known as the insurer's ceded and retained loss functions, respectively. To reduce ex post moral hazard, we follow Huberman et al. (1983) to assume that both the insurer and the reinsurer are obligated to pay more for a larger loss in a typical reinsurance treaty. In other words, both ceded and retained loss functions are constrained to be nondecreasing, and we use $\mathfrak{C}$ to denote the set of admissible ceded loss functions:

$$\mathfrak{C} \triangleq \{0 \leq f(x) \leq x : \text{both } R_f(x) \text{ and } f(x) \text{ are nondecreasing functions}\}. \tag{2.1}$$

The ceded loss functions in the above set have many nice properties. For instance, as shown in Chi and Tan (2011), $f(x) \in \mathfrak{C}$ is nondecreasing and Lipschitz continuous,

$$0 \leq f(x_2) - f(x_1) \leq x_2 - x_1, \quad \forall\, 0 \leq x_1 \leq x_2, \tag{2.2}$$

which is equivalent to $f'(x) \in [0, 1]$, *a.s.* where $f'(x)$ represents the derivative of $f(x)$.

Because part of loss is ceded to a reinsurer, the insurer in a reinsurance arrangement incurs an additional cost in the form of the reinsurance premium, which is payable to the reinsurer. It is well known that the reinsurance market is incomplete. Thus, Bühlmann (1980) advocates that the reinsurance premium should be calculated based not only on the ceded loss but also on the market environment. For simplicity, we use a non-negative random variable $\mathcal{M}$ with $\mathbb{E}[\mathcal{M}] = 1$ to reflect the reinsurance market environment and other risks the reinsurer is facing, and let the reinsurance premium be calculated by an economic premium principle as

$$\pi_{\mathcal{M}}(Y) = \mathbb{E}[Y\mathcal{M}] \tag{2.3}$$

for all non-negative random variable Y. From the above equation, it is easy to see that the reinsurance premium is no longer law-invariant and relies heavily on the joint distribution of the ceded loss and the market factor. Moreover, $\pi_{\mathcal{M}}(.)$ typically does not preserve the convex order. Thus, such a premium principle is very different from that considered in Chi (2012). Indeed, this class of premium principles is quite general and includes many widely used premium principles such as weighted premium principles. More specifically, the economic premium principle becomes the weighted premium principle of Furman and Zitikis (2009) by setting the market factor to be

$$\mathcal{M} = \frac{w(Z)}{\mathbb{E}[w(Z)]} \tag{2.4}$$

for a non-negative random variable Z and a non-negative nondecreasing function $w(.)$, provided that the expectation exists. By an appropriate specification of the weighting function $w(.)$, the weighted premium principle in turn recovers many popular premium principles; these include net, modified covariance, size-biased, Esscher, Aumann-Shapley, Kamps, excess-of-loss, distorted, proportional hazard, CTE, and modified tail covariance premium principles, as shown in Table 1, which is reproduced from Table 2 of Furman and Zitikis (2009) for convenience. Note that the Esscher premium principle is also known as the exponential tilting in Wang (2002) even if Z is negative.

On the selection of market factors that can be integrated into the premium principle, Zhu et al. (2016) provide an in-depth analysis. In particular, using some Canadian empirical agricultural data, they demonstrate that the Canadian GDP can be a useful market factor for calculating the agricultural reinsurance premium.

In the presence of reinsurance, the liability or the total risk exposure of the insurer is now given by the sum of the retained loss and the incurred reinsurance premium instead of X. Using $T_f(X)$ to denote the resulting liability of the insurer, we have

$$T_f(X) = R_f(X) + \pi_{\mathcal{M}}(f(X)). \tag{2.5}$$

To evaluate the liability of the insurer, we use a cost-of-capital approach that was recently introduced by Swiss Federal Office of Private Insurance (2006) and has been widely adopted by the insurance companies in Europe. Specifically, the best estimate of the insurer's liability is usually represented by $\mathbb{E}[T_f(X)]$. From a regulatory risk capital requirement point of view, it is prudent to also

TABLE 1
Some Weighted Premium Principles

Weighted premium principle	$w(z)$	$\Pi_{\mathcal{M}}(Y)$
Net	Constant	$\mathbb{E}[Y]$
Modified covariance	z	$\mathbb{E}[Y] + Cov[Z, Y]/\mathbb{E}[Z]$
Size-biased	z^t	$\mathbb{E}[YZ^t]/\mathbb{E}[Z^t]$
Esscher	e^{tz}	$\mathbb{E}[Ye^{tZ}]/\mathbb{E}[e^{tZ}]$
Aumann-Shapley	$e^{tF_Z(z)}$	$\mathbb{E}[Ye^{tF_Z(Z)}]/\mathbb{E}[e^{tF_Z(Z)}]$
Kamps	$1 - e^{-tz}$	$\mathbb{E}[Y(1 - e^{-tZ})]/\mathbb{E}[(1 - e^{-tZ})]$
Excess-of-loss	$\mathbb{I}_{\{z\geq t\}}$	$\mathbb{E}[Y\|Z \geq t]$
Distorted	$g'(S_Z(z))$	$\mathbb{E}[Yg'(S_Z(Z))]$
Proportional hazard	$pS_Z^{p-1}(z)$	$p\mathbb{E}[YS_Z^{p-1}(Z)]$
CTE	$\mathbb{I}_{\{z\geq VaR_p(Z)\}}$	$\mathbb{E}[Y\|Z \geq VaR_p(Z)]$
Modified tail covariance	$z\mathbb{I}_{\{z\geq VaR_p(Z)\}}$	$\mathbb{E}[YZ\|Z \geq VaR_p(Z)]/\mathbb{E}[Z\|Z \geq VaR_p(Z)]$

Here $t \in [0, \infty)$, $p \in [0, 1)$, $S_Z(z)$ is the survival function of a random variable Z, and $\mathbb{I}_{\mathcal{A}}$ is the indicator function of an event $\mathcal{A}$. $g(.)$ is a nondecreasing concave distortion function with $g(0) = 0$ and $g(1) = 1$, and $VaR_p(Z)$, representing the VaR of Z at the confidence level $1 - p$, is formally defined in Definition 2.1.

account for the unexpected loss $T_f(X) - \mathbb{E}[T_f(X)]$. If the risk measure φ is used to quantify the risk associated with the unexpected loss, then the capital at risk corresponds to

$$\varphi(T_f(X) - \mathbb{E}[T_f(X)]).$$

In practice, the return from the capital investment is much smaller than that required by shareholders. By denoting $\delta \in (0, 1)$ as the return difference, which is known as the cost-of-capital rate, the risk margin is then given by the product of the cost-of-capital rate and the capital at risk. The risk-adjusted value or market-consistent price of the insurer's liability, $\mathcal{L}_f^{\varphi}(X)$, is defined as

$$\mathcal{L}_f^{\varphi}(X) = \mathbb{E}[T_f(X)] + \delta\varphi(T_f(X) - \mathbb{E}[T_f(X)]). \tag{2.6}$$

For more details on evaluating the insurer's liability with a cost-of-capital approach, see Swiss Federal Office of Private Insurance (2006) and Risk Margin Working Group (2009).

In the insurance and other financial industry, VaR and CVaR risk measures have been widely used for quantifying risk and setting regulatory capital. Formally, these two risk measures are defined as follows:

Definition 2.1. The VaR of a random variable Y at a confidence level $1 - \alpha$ where $0 < \alpha < 1$ is defined as

$$VaR_\alpha(Y) \triangleq \inf\{t \in \mathbb{R} : S_Y(t) \leq \alpha\}, \tag{2.7}$$

where $\inf \emptyset = \infty$ by convention. Based upon the definition of VaR, CVaR of Y at the confidence level $1 - \alpha$ is defined as

$$CVaR_\alpha(Y) \triangleq \frac{1}{\alpha}\int_0^\alpha VaR_s(Y)\, ds. \tag{2.8}$$

It follows from the definition of VaR that

$$VaR_\alpha(Y) \leq z \Longleftrightarrow S_Y(z) \leq \alpha \tag{2.9}$$

holds for any $z \in \mathbb{R}$. Therefore, for the non-negative random variable X, we have $VaR_\alpha(X) = 0$ for $\alpha \geq S_X(0)$. For this reason, we assume in this article that the parameter α satisfies $0 < \alpha < S_X(0)$ to avoid trivial cases. Another important property associated

with $VaR_\alpha(Y)$ is that for any nondecreasing continuous function $m(x)$, we have (Dhaene et al. 2002, Theorem 1)

$$VaR_\alpha(m(Y)) = m\left(VaR_\alpha(Y)\right). \tag{2.10}$$

The risk measure CVaR is also known as the "average value-at-risk" and the "expected shortfall." A key advantage of CVaR over VaR is that the former is a coherent risk measure whereas the latter is not because it fails to satisfy the subadditivity property. More detailed discussions on the properties of VaR and CVaR risk measures can be found in Artzner et al. (1999) and Föllmer and Schied (2004).

By specifying φ to be either VaR or CVaR risk measure at the confidence level $1-\alpha$, the respective optimal reinsurance models can be formulated as follows:

$$\text{VaR-related optimization}: \quad \mathcal{L}_{f^*}^{VaR_\alpha}(X) = \min_{f\in\mathfrak{C}} \mathcal{L}_f^{VaR_\alpha}(X) \tag{2.11}$$

and

$$\text{CVaR-related optimization}: \quad \mathcal{L}_{f^*}^{CVaR_\alpha}(X) = \min_{f\in\mathfrak{C}} \mathcal{L}_f^{CVaR_\alpha}(X), \tag{2.12}$$

where f^* is the resulting optimal ceded loss function among $\mathfrak{C}$. The rest of this article is devoted to analyzing the solutions to these two optimal reinsurance models assuming that the reinsurance premium is calculated by an economic premium principle.

3. OPTIMAL REINSURANCE POLICIES

In this section, we solve the optimal reinsurance models as formulated in (2.11) and (2.12). As pointed out earlier, one of the key features of our optimal reinsurance models is that the reinsurance premium is determined by an economic premium principle, which in turn takes into consideration the market factor $\mathcal{M}$. Here we assume that the weighted c.d.f. of X with respect to the market factor $\mathcal{M}$ is represented by

$$F_X^{\mathcal{M}}(t) \triangleq \mathbb{E}[\mathcal{M} \times \mathbb{I}_{\{X\le t\}}], \quad \forall t \ge 0, \tag{3.1}$$

with its corresponding survival function $S_X^{\mathcal{M}}(t) \triangleq 1 - F_X^{\mathcal{M}}(t)$, for any $t \ge 0$. This notation can be used to provide another expression of reinsurance premium. More specifically, for any admissible ceded risk $f(X)$ and noting $f(x)$ is a nondecreasing Lipschitz-continuous function with $f(0) = 0$, it follows from Fubini's theorem that

$$\pi_{\mathcal{M}}(f(X)) = \mathbb{E}\left[\mathcal{M} \times \int_0^\infty \mathbb{I}_{\{X>t\}}\, df(t)\right] = \int_0^\infty S_X^{\mathcal{M}}(t)\, df(t). \tag{3.2}$$

Theorem 3.1 below solves the VaR-related reinsurance model (2.11). The proof to this theorem can be found in the Appendix.

Theorem 3.1. *An optimal solution to the VaR-related reinsurance model (2.11) is given by*

$$f_v^*(x) = \int_0^x \mathbb{I}_{\{\phi(t)\le 0\}}\, dt, \tag{3.3}$$

where

$$\phi(t) \triangleq S_X^{\mathcal{M}}(t) - (1-\delta)S_X(t) - \delta\mathbb{I}_{\{S_X(t)>\alpha\}}, \quad \forall t \ge 0. \tag{3.4}$$

Note that the optimal ceded loss function (3.3) is expressed in the form of an integration over an indicator function. Although the optimal ceded loss function is fully determined by the sign of the function $\phi(t)$, $\phi(t)$ is known only to be right-continuous, and its monotonicity, which relies heavily on the dependence between $\mathcal{M}$ and X, is unclear.

The optimal reinsurance under CVaR risk measure can be derived analogously and is formally stated in the following theorem.

Theorem 3.2. *A ceded loss function that optimally solves the CVaR-related reinsurance problem (2.12) is given by*

$$f_{cv}^*(x) = \int_0^x \mathbb{I}_{\{\psi(t)<0\}}\, dt, \tag{3.5}$$

where

$$\psi(t) \triangleq S_X^{\mathcal{M}}(t) - (1-\delta)S_X(t) - \delta \min\{1, S_X(t)/\alpha\}. \tag{3.6}$$

The proof to the above theorem is relegated to the Appendix. Here we note the following:

- First, it is of interest to note that from Theorems 3.1 and 3.2, the piecewise linear reinsurance treaty is always optimal regardless of which risk measure, VaR or CVaR, is adopted. Second, both optimal ceded loss functions differ by the terms $\mathbb{I}_{\{S_X(t)>\alpha\}}$ and $\min\{1, S_X(t)/\alpha\}$. Third, it is easy to see that

$$\mathbb{I}_{\{S_X(t)>\alpha\}} \le \min\{1, S_X(t)/\alpha\}, \quad \forall t \ge 0,$$

  with equality for $0 \le t < VaR_\alpha(X)$ or $t \ge Supp(X)$, where $Supp(X) \triangleq \inf\{t \ge 0 : \mathbb{P}(X > t) = 0\}$. Thus, it follows from (3.3) and (3.5) that $f_v^*(x) = f_{cv}^*(x)$ for any $x \in [0, VaR_\alpha(X)]$ and $f_v^*(x) \le f_{cv}^*(x)$ for each $x > VaR_\alpha(X)$. In other words, if the regulatory risk measure is changed from VaR to CVaR, the impact is for the insurer to cede more tail loss at the same confidence level. This is intuitive since CVaR, as opposed to VaR, demands more regulatory capital on risk. This finding is also consistent with that obtained by Chi (2012) under the expected value principle and Wang's premium principle.
- By further comparing our results with Chi (2012), we can see that optimal reinsurance under the economic reinsurance premium principle appears to be more complicated than that under most of the widely used premium principles listed in Young (2004). More specifically, Chi (2012) shows that it is optimal for the insurer to purchase a one-layer reinsurance of the form

$$f^*(x) = \int_0^x \mathbb{I}_{\{t\in(a^*,b^*]\}}\, dt, \quad \forall x \ge 0$$

  for some $0 \le a^* \le b^*$. The optimality of one-layer reinsurance is also obtained by Cong and Tan (2016), who consider minimizing the insurer's risk exposure under VaR by assuming that the reinsurance premium principle is piecewise and monotonic.
- The piecewise linear optimal reinsurance policies under both VaR and CVaR differ only in how the tail risk is ceded. The difference is further highlighted in their effect by the cost-of-capital rate δ. More specifically, it is easy to see that $\psi(t)$ is nonincreasing in δ, and this in turn implies that the insurer would buy more reinsurance under CVaR when the capital becomes more costly. On the other hand, although $\phi(t)$ is also nonincreasing in δ for $t \le VaR_\alpha(X)$, it becomes nondecreasing for each $t > VaR_\alpha(X)$. The consequence is that under VaR it may be optimal for the insurer to retain more tail risk for a higher cost-of-capital rate.
- It is noteworthy that Balbás et al. (2015) consider a quite general reinsurance model, which includes our CVaR-related reinsurance model as a special case. Although they conclude that the optimal contract is often a bang-bang solution, few explicit optimal solutions are obtained in their work. In contrast, in this article we derive an explicit form of the optimal reinsurance. This facilitates the comparative analysis, which in turn provides greater insight on the impact of some model factors to our proposed optimal reinsurance.

4. THE EFFECTS OF THE MARKET FACTOR ON OPTIMAL REINSURANCE

In addition to the critical role of the underlying loss distribution of X in determining the optimal reinsurance policies (3.3) and (3.5), the market factor $\mathcal{M}$ also has a nontrivial effect on the optimality, as reflected through the weighted survival function $S_X^{\mathcal{M}}(t)$. Thus, it is of interest to study its impact on the optimal reinsurance policies. In particular, in this section we will analyze the effects of the market factor on optimal reinsurance design from two aspects: (1) special optimal reinsurance is derived by assuming some stochastic dependence between the market factor $\mathcal{M}$ and the loss X (Subsection 4.1) and (2) optimal reinsurance policies are compared for different market factors (Subsection 4.2).

4.1. Expectation Dependence

This subsection analyzes the effect of stochastic dependence between the market factor $\mathcal{M}$ and the loss X on the optimal reinsurance design under both VaR and CVaR risk measures. To facilitate our analysis, the stochastic dependence is expressed in the form of the expectation dependence, which is defined formally as follows.

Definition 4.1. A random variable Y_1 is said to be positively expectation dependent (PED) with respect to a random variable Y_2 if

$$\mathbb{E}[Y_1|Y_2 \leq y] \leq \mathbb{E}[Y_1], \quad \forall y \in \mathbb{R}.$$

Similarly, Y_1 is said to be negatively expectation dependent (NED) with respect to Y_2 if

$$\mathbb{E}[Y_1|Y_2 \leq y] \geq \mathbb{E}[Y_1], \quad \forall y \in \mathbb{R}.$$

The above two notions of expectation dependence are attributed to Wright (1987). The average estimate of Y_1 depends on the relative magnitude of Y_2. In particular, when we discover that Y_2 is smaller, or more precisely $Y_2 \leq y$, the (conditional) expectation of Y_1 is adjusted downward for PED but upward for NED. These forms of stochastic dependence have been applied by Wright (1987) to analyze a risk averse investor's portfolio investment decision. These notions of stochastic dependence have also been used by Hong et al. (2011) to generalize Mossin's Theorem to a setting with random initial wealth W. Given a fair premium, Hong et al. (2011) establish that any risk averse individual purchases less than or more than full insurance if and only if the random loss X is PED or NED with respect to random initial wealth, respectively.

Here we similarly use the above stochastic dependence to analyze its impact on the optimal decision to cede losses. We show that the resulting optimal ceded loss functions have very simple structures, as formally established in the following proposition and Proposition 4.2 for optimal reinsurance models (2.12) and (2.11), respectively. Their proofs are collected in the Appendix.

Proposition 4.1. *If the market factor $\mathcal{M}$ is NED with respect to the loss X, then the optimal solution to the CVaR-related reinsurance model (2.12) is full reinsurance in that the entire loss is transferred to the reinsurer.*

Proposition 4.2. *For the VaR-related reinsurance model (2.11), if $\mathcal{M}$ is PED with respect to X, then it is optimal for the insurer to retain the risk over $VaR_\alpha(X)$ with $f_v^*(x) = f_v^*(VaR_\alpha(X))$ for all $x \geq VaR_\alpha(X)$. On the other hand, if $\mathcal{M}$ is NED with respect to X, then the insurer will cede all the risk less than $VaR_\alpha(X)$ with $f_v^*(VaR_\alpha(X)) = VaR_\alpha(X)$.*

Proposition 4.1 is very intuitive once we understand the implication of $\mathcal{M}$ being NED with respect to X. More precisely, when $\mathcal{M}$ is NED with respect to X, it follows from (3.2) and (A.5) that

$$\pi_{\mathcal{M}}(f(X)) \leq \mathbb{E}[f(X)]$$

for any admissible ceded loss function $f(x) \in \mathfrak{C}$. Hence the reinsurance premium is less than the net premium, and thus it is not surprising that full reinsurance is optimal. This result is also consistent with Mossin's Theorem.

Comparing both Propositions 4.1 and 4.2, the effect of the stochastic dependence on the optimal reinsurance design can be different depending on the choice of the risk measure. In particular, when the market factor is NED with respect to the loss, the optimal solution under VaR risk measure is full reinsurance only if

$$S_X^{\mathcal{M}}(t) \leq (1-\delta)S_X(t), \quad \forall t \in [VaR_\alpha(X), Supp(X)).$$

Obviously, this condition is not always satisfied. For instance, when X is independent of $\mathcal{M}$, we have $S_X^{\mathcal{M}}(t) = S_X(t)$, and hence the above condition fails to be satisfied. For this case, the optimal ceded loss function under VaR risk measure is $f_v^*(x) = \min\{x, VaR_\alpha(X)\}$.

4.2. Comparative Analysis

In this subsection, we provide a comparative analysis of the effects of different market factors on the optimal reinsurance policies. We analyze these effects through a copula. The copula function, which is formally defined below, can be a very useful tool to capture the stochastic dependence of random vectors.

Definition 4.2. A bivariate copula is a function $C(u, v)$ with domain $[0, 1]^2$ satisfying the following properties:

1. For any $u, v \in [0, 1]$,

$$C(u, 0) = C(0, v) = 0 \quad \text{and} \quad C(u, 1) = u, \quad C(1, v) = v; \tag{4.1}$$

2. $C(u, v)$ is 2-increasing, that is,

$$C(u_2, v_2) - C(u_1, v_2) - C(u_2, v_1) + C(u_1, v_1) \geq 0$$

for any $0 \leq u_1 \leq u_2 \leq 1$ and $0 \leq v_1 \leq v_2 \leq 1$.

By Sklar's Theorem, there exists a copula $C_{\mathcal{M},X}(u, v)$ such that

$$\mathbb{P}(\mathcal{M} \leq t, X \leq x) = C_{\mathcal{M},X}\left(F_{\mathcal{M}}(t), F_X(x)\right)$$

for any $(t, x) \in \mathbb{R}^2$. See Nelsen (2006) for detailed discussions on copulas.

For our first set of comparative analysis, we examine the optimal reinsurance policies for different copulas of $(\mathcal{M}, X)$ by fixing the distribution function of the market factor. The results are summarized in the following proposition.

Proposition 4.3. *For two market factors $\mathcal{M}_1$ and $\mathcal{M}_2$ with the same distribution function $F_{\mathcal{M}}(t)$, if $C_{\mathcal{M}_1,X}(u, v) \geq C_{\mathcal{M}_2,X}(u, v)$ for any $(u, v) \in [0, 1]^2$, then the insurer would cede less risk under $\mathcal{M}_1$ than that under $\mathcal{M}_2$ for both VaR and CVaR risk measures.*

The result established in the above proposition is intuitive. As shown in the proof of the above proposition in the Appendix, if $C_{\mathcal{M}_1,X}(u, v) \geq C_{\mathcal{M}_2,X}(u, v)$ for any $(u, v) \in [0, 1]^2$, we must have $S_X^{\mathcal{M}_1}(t) \geq S_X^{\mathcal{M}_2}(t)$, $\forall t \geq 0$. This result, together with (3.2) and the assumptions in the above proposition, implies that the reinsurance premium is higher under the market factor $\mathcal{M}_1$ than that under $\mathcal{M}_2$. Thus, it is intuitive that the insurer would cede less risk when the reinsurance is more costly.

Next, we compare optimal reinsurance design for different distribution functions of market factors by fixing the copula of the market factor and the loss X. The notion of "convex order," as formally defined below, will facilitate the comparison of distribution functions.

Definition 4.3. A random variable Y_1 is said to be smaller than Y_2 in the sense of convex order (denoted by $Y_1 \leq_{cx} Y_2$) if

$$\mathbb{E}[Y_1] = \mathbb{E}[Y_2] \quad \text{and} \quad \mathbb{E}[(Y_1 - d)_+] \leq \mathbb{E}[(Y_2 - d)_+], \quad \forall d \in \mathbb{R},$$

provided that the expectations exist, where $(x)_+ = \max(x, 0)$.

For more details on stochastic orders, we refer to Müller and Scarsini (2001) and Shaked and Shanthikumar (2007).

By imposing a weak condition on the copula, the comparison of optimal reinsurance for different distributions of the market factor is given in the following proposition. Its proof is provided in the Appendix.

Proposition 4.4. *For two market factors $\mathcal{M}_1$ and $\mathcal{M}_2$, assume that $(\mathcal{M}_i, X)$ for $i = 1, 2$ have the same copula $C(u, v)$ and $\mathcal{M}_1 \leq_{cx} \mathcal{M}_2$. If $\frac{\partial C(u,v)}{\partial u}$ is nonincreasing in u for every $v \in [0, 1]$, then we have $S_X^{\mathcal{M}_1}(t) \leq S_X^{\mathcal{M}_2}(t)$ for any $t \geq 0$ such that the insurer would cede more risk under $\mathcal{M}_1$ than that under $\mathcal{M}_2$; otherwise, if $\frac{\partial C(u,v)}{\partial u}$ is nondecreasing in u for every $v \in [0, 1]$, then the opposite conclusion is true.*

Recall that the economic premium principle is a rather general premium principle and encompasses many widely used premium principles as special cases. In particular, depending on the specification of the weighting function of the market factor, we can recover at least 11 premium principles as listed in Table 1. Most of these weighting functions are determined by a parameter, which is denoted by either t or p. Thus, it is of interest to analyze the effects of the change of these parameters on the optimal reinsurance design, which are expected to heavily rely on the dependence between X and Z since $\mathcal{M}$ is a function of Z in (2.4) and the above proposition shows that the relationship between $\mathcal{M}$ and X plays a critical role in the insurer's risk transfer. These results are summarized in the following corollary with its proof relegated to the Appendix.

Corollary 4.1. *Under both VaR and CVaR risk measures, if the loss X is stochastically increasing with respect to Z, which is denoted by $X \uparrow_{SI} Z$, then for a larger premium principle parameter, the insurer would cede less risk for Esscher, size-biased, excess-of-loss or Aumann-Shapley premium principle but retain less risk for Kamps, proportional hazard, CTE, or modified tail covariance premium principle.*[1] *Otherwise, the opposite results hold if $X \uparrow_{SI} -Z$.*

Let $C(u, v)$ be the copula function for the random vector (Z, X). If $\frac{\partial C(u,v)}{\partial u}$ is nonincreasing in u for each $v \in [0, 1]$, then it follows from Proposition 2.12 in Cai and Wei (2012) that $X \uparrow_{SI} Z$, which in turn implies $X \uparrow_{SI} \mathcal{M}$. Otherwise, if it becomes nondecreasing, then we have $X \uparrow_{SI} -Z$. Further, we can see from (3.2) and the proof of the above corollary that if $X \uparrow_{SI} Z$, then the reinsurance becomes more costly for Esscher, size-biased, excess-of-loss, and Aumann-Shapley premium principles with a larger parameter, and for Kamps, proportional hazard, CTE, and modified tail covariance premium principles with a smaller parameter.

5. OPTIMAL REINSURANCE FOR BIVARIATE COPULAS WITH QUADRATIC SECTIONS

Motivated by the critical role of the bivariate copula of the market factor $\mathcal{M}$ and the loss X in the optimal reinsurance design, this section derives explicitly the optimal ceded loss functions under the additional assumption of bivariate copula with quadratic sections. Formally, a bivariate copula with quadratic sections is represented by

$$C(u, v) = uv + \lambda(u)v(1 - v), \tag{5.1}$$

where $\lambda(u)$ is a function over $[0, 1]$. According to Theorem 3.2.4 in Nelsen (2006), $C(u, v)$ defined above is a copula if and only if $\lambda(u)$ satisfies $\lambda(0) = \lambda(1) = 0$ and

$$|\lambda(u_1) - \lambda(u_2)| \le |u_1 - u_2|, \quad \forall 0 \le u_1, u_2 \le 1. \tag{5.2}$$

In the special case where $\lambda(u) = \theta u(1 - u)$ for a $\theta \in [-1, 1]$, the above copula recovers the well-known Farlie-Gumbel-Morgenstern (FGM) copula.

By means of a copula with quadratic sections, we can characterize the expectation dependence of $\mathcal{M}$ with respect to the loss X. This is demonstrated in the following proposition with its proof relegated to the Appendix.

Proposition 5.1. *If the market factor $\mathcal{M}$ and the loss X have a bivariate copula of the form (5.1), then $\mathcal{M}$ is PED (NED) with respect to X if and only if $Q_\lambda^{\mathcal{M}}$ is non-negative (non-positive), where*

$$Q_\lambda^{\mathcal{M}} \triangleq \int_0^\infty \lambda(F_{\mathcal{M}}(y))\, dy \in [-1, 1]. \tag{5.3}$$

In addition, armed with this assumption, we are able to show that the resulting optimal ceded loss functions have relatively simple structures, as presented in the following proposition. The proof is given in the Appendix.

Proposition 5.2. *Under the same assumption of Proposition 5.1 and that $\delta \ge \alpha$, the optimal ceded loss functions of the VaR-related and CVaR-related reinsurance models (2.11) and (2.12) are given, respectively, by*

$$f_v^*(x) = \begin{cases} \min\{x, VaR_\alpha(X)\} + \left(x - VaR_{\min\{\alpha, 1+\frac{\delta}{Q_\lambda^{\mathcal{M}}}\}}(X)\right)_+, & Q_\lambda^{\mathcal{M}} \in [-1, -\delta); \\ \min\{x, VaR_\alpha(X)\}, & Q_\lambda^{\mathcal{M}} \in [-\delta, \delta]; \\ \min\left\{\left(x - VaR_{\frac{\delta}{Q_\lambda^{\mathcal{M}}}}(X)\right)_+, VaR_\alpha(X) - VaR_{\frac{\delta}{Q_\lambda^{\mathcal{M}}}}(X)\right\}, & Q_\lambda^{\mathcal{M}} \in (\delta, 1] \end{cases}$$

[1] A random variable Y_1 is said to be stochastically increasing with respect to a random variable if $\mathbb{P}(Y_1 > y_1 | Y_2 = y_2)$ is nondecreasing over the support of Y_2 for any $y_1 \in \mathbb{R}$. See Shaked and Shanthikumar (2007) for more details.

and

$$f_{cv}^{*}(x)=\begin{cases} x, & Q_{\lambda}^{\mathcal{M}}\in[-1,0];\\ \min\left\{\left(x-VaR_{\frac{\delta}{Q_{\lambda}^{\mathcal{M}}}}(X)\right)_{+}, VaR_{1-\frac{\delta}{Q_{\lambda}^{\mathcal{M}}}(1/\alpha-1)}(X)-VaR_{\frac{\delta}{Q_{\lambda}^{\mathcal{M}}}}(X)\right\}, & \text{otherwise.}\end{cases}$$

When the market factor $\mathcal{M}$ and the loss X have a copula with quadratic sections, the above proposition asserts that the optimal contract is either one-layer or two-layer. Furthermore, the insurer's optimal risk transfer is affected by the market factor $\mathcal{M}$ through $Q_{\lambda}^{\mathcal{M}}$, and the insurer would cede less risk for a larger $Q_{\lambda}^{\mathcal{M}}$ under both VaR and CVaR risk measures. Moreover, ceding the very tail risk for $Q_{\lambda}^{\mathcal{M}}\in[-1,-\delta)$ can be rather surprising since under the VaR risk measure it is generally optimal to retain the tail risk. Finally, it should be pointed out that $\delta\geq\alpha$ is a reasonable assumption in practice.

To conclude this section, we present the following example to highlight the role of the market factor on the reinsurance design.

Example 5.1. We begin the example by assuming that the market factor $\mathcal{M}$ can be captured by the following exponential tilting function (Wang 2002):

$$\mathcal{M}=e^{\eta N}/\mathbb{E}[e^{\eta N}]$$

for a constant $\eta\geq 0$, and without loss of generality, N is a standardized normal random variable. We further assume that the stochastic dependence between the market factor $\mathcal{M}$ and the loss X can be modeled by the FGM copula so that $Q_{\lambda}^{\mathcal{M}}$ in (5.3) simplifies to

$$Q_{\lambda}^{\mathcal{M}}=\theta\left(1-2\mathcal{N}\left(-\frac{\eta}{\sqrt{2}}\right)\right) \tag{5.4}$$

for $\theta\in[-1,1]$, and $\mathcal{N}(\cdot)$ is the c.d.f. of standard normal distribution. See the Appendix for the proof of this result.

If $\eta=0$, then it follows from (5.4) that $Q_{\lambda}^{\mathcal{M}}=0$ so that Proposition 5.2 yields $f_{v}^{*}(x)=\min\{x, VaR_{\alpha}(X)\}$ and $f_{cv}^{*}(x)=x$.

Otherwise, if $\eta>0$, then it can be seen from (5.4) that $Q_{\lambda}^{\mathcal{M}}$ is an increasing function in θ. Note that θ reflects the degree of dependence between the market factor $\mathcal{M}$ and the loss X and that the Kendall's tau equals to $2\theta/9$. Thus, the above argument implies that the insurer cedes less risk for a larger Kendall's tau. Moreover, $Q_{\lambda}^{\mathcal{M}}>0$ if and only if $\theta>0$. When $\mathcal{M}$ and X are positively dependent, $Q_{\lambda}^{\mathcal{M}}$ is increasing in η such that the insurer will cede less risk for a larger η. Otherwise, if they are negatively dependent, a similar argument implies that the ceded risk increases in η under both VaR and CVaR risk measures.

6. CONCLUSION

In this article, we push the boundary on the research of optimal reinsurance by considering a reinsurance model that minimizes the risk-adjusted value of an insurer's liability subject to the reinsurance premium being calculated by an economic premium principle and that both insurer and reinsurer are obligated to pay more for a larger realization of loss. Here the valuation of the insurer's liability is based on a cost-of-capital approach with the capital at risk being quantified by a VaR or CVaR risk measure, as recommended by Swiss Federal Office of Private Insurance (2006) and Risk Margin Working Group (2009).

Although the underlying formulation of the optimal reinsurance model has appeared in the literature (see Chi 2012; Asimit et al. 2013), the assumption of an economic premium principle on reinsurance pricing is, new and important. In addition to the desirable property that it is a rather general class of premium principles that encompasses many popular premium principles as special cases, more importantly the economic premium principle has the capability of integrating the market factor into the reinsurance pricing framework. As the reinsurance market is generally perceived to be incomplete, in practice the reinsurance premium depends not only on the loss ceded but also on the market economic factor. The economic premium principle is able to reflect this feature and hence is more consistent with market practice.

Because of the integration of the market economic factor (via the reinsurance pricing) into the optimal reinsurance model, some new insights on the optimal reinsurance design could be gleaned, which would otherwise be impossible for the existing models. For example, it was formally shown that the optimality is greatly influenced by the changes of the stochastic dependence between the market factor and the loss. A comparative study was also conducted to investigate the effect of the market factor on the optimal reinsurance policies. Finally, it was shown that optimal reinsurance policies could admit relatively simple forms under an additional assumption that the market factor and the loss have a copula with quadratic sections.

FUNDING

All three authors greatly acknowledge a research grant from the Society of Actuaries Committee on Knowledge Extension Research. Chi's research was also supported by the National Natural Science Foundation of China (Grant No. 11471345). Tan is also grateful for support from the SOA CAE and the Natural Sciences and Engineering Research Council of Canada (pin# 220010).

REFERENCES

Arrow, K. J. 1963. Uncertainty and the Welfare Economics of Medical Care. *American Economic Review* 53(5): 941–973.

Artzner, P., F. Delbaen, J. M. Eber, and D. Heath, 1999. Coherent Measures of Risk. *Mathematical Finance* 9(3): 203–228.

Asimit, A. V., A. M. Badescu, and A. Tsanakas, 2013. Optimal Risk Transfers in Insurance Groups. *European Actuarial Journal* 3(1): 159–190.

Balbás, A., B. Balbás, and A. Heras, 2009. Optimal Reinsurance with General Risk Measures. *Insurance: Mathematics and Economics* 44(3): 374–384.

Balbás, A., B. Balbás, R. Balbás, and A. Heras, 2015. Optimal Reinsurance under Risk and Uncertainty. *Insurance: Mathematics and Economics* 60: 61–74.

Bernard, C., and W. Tian, 2009. Optimal Reinsurance Arrangements under Tail Risk Measures. *Journal of Risk and Insurance* 76(3): 709–725.

Borch, K. 1960. An Attempt to Determine the Optimum Amount of Stop Loss Reinsurance. In *Transactions of the 16th International Congress of Actuaries*, vol. I, 597–610. Brussels, Belgium: Georges Thone.

Bühlmann, H. 1980. An Economic Premium Principle. *Astin Bulletin* 11(1): 52–60.

Cai, J., and K. S. Tan, 2007. Optimal Retention for a Stop-Loss Reinsurance under the VaR and CTE Risk Measures. *Astin Bulletin* 37(1): 93–112.

Cai, J., K. S. Tan, C. Weng, and Y. Zhang, 2008. Optimal Reinsurance under VaR and CTE Risk Measures. *Insurance: Mathematics and Economics* 43(1): 185–196.

Cai, J., and W. Wei, 2012. On the Invariant Properties of Notions of Positive Dependence and Copulas under Increasing Transformations. *Insurance: Mathematics and Economics* 50(1): 43–49.

Cheung, K. C. 2010. Optimal Reinsurance Revisited—A Geometric Approach. *Astin Bulletin* 40(1): 221–239.

Chi, Y. 2012. Reinsurance Arrangements Minimizing the Risk-Adjusted Value of an Insurer's Liability. *Astin Bulletin* 42(2): 529–557.

Chi, Y., and K. S. Tan, 2011. Optimal Reinsurance under VaR and CVaR Risk Measures: A Simplified Approach. *Astin Bulletin* 41(2): 487–509.

Chi, Y., and K. S. Tan, 2013. Optimal Reinsurance with General Premium Principles. *Insurance: Mathematics and Economics* 52(2): 180–189.

Chi, Y., and X. S. Lin, 2014. Optimal Reinsurance with Limited Ceded Risk: A Stochastic Dominance Approach. *Astin Bulletin* 44(1): 103–126.

Cong, J., and K. S. Tan, 2016. Optimal VaR-Based Risk Management with Reinsurance. *Annals of Operations Research* 237(1): 177–202.

Dhaene, J., M. Denuit, M. J. Goovaerts, R. Kaas, and D. Vyncke, 2002. The Concept of Comonotonicity in Actuarial Science and Finance: Theory. *Insurance: Mathematics and Economics* 31(1): 3–33.

Föllmer, H., and A. Schied, 2004. *Stochastic Finance: An Introduction in Discrete Time*. 2nd rev. ed. New York: Walter de Gruyter.

Furman, E., and R. Zitikis, 2009. Weighted Pricing Functionals with Applications to Insurance: An Overview. *North American Actuarial Journal* 13(4): 483–496.

Hong, S. K., K. O. Lew, R. MacMinn, and P. Brockett, 2011. Mossin's Theorem given Random Initial Weath. *Journal of Risk and Insurance* 78(2): 309–324.

Huberman, G., D. Mayers, and C. W. Smith Jr. 1983. Optimal Insurance Policy Indemnity Schedules. *Bell Journal of Economics* 14(2): 415–426.

Kaluszka, M. 2001. Optimal Reinsurance under Mean-Variance Premium Principles. *Insurance: Mathematics and Economics* 28(1): 61–67.

Kaluszka, M., and A. Okolewski, 2008. An Extension of Arrow's Result on Optimal Reinsurance Contract. *The Journal of Risk and Insurance* 75(2): 275–288.

Merz, M., and M. V. Wüthrich, 2014. Demand of Insurance under the Cost-of-Capital Premium Calculation Principle. *Risks* 2(2): 226–248.

Müller, A., and M. Scarsini, 2001. Stochastic Comparison of Random Vectors with a Common Copula. *Mathematics of Operations Research* 26(4): 723–740.

Nelsen, R. B. 2006. *An Introduction to Copulas*. 2nd ed. New York: Springer.

Ohlin, J. 1969. On a Class of Measures of Dispersion with Application to Optimal Reinsurance. *Astin Bulletin* 5(2): 249–266.

Raviv, A. 1979. The Design of an Optimal Insurance Policy. *American Economic Review* 69(1): 84–96.

Risk Margin Working Group. 2009. Measurement of Liabilities for Insurance Contracts: Current Estimates and Risk Margins. *City: International Actuarial Association*. Ottawa, Canada.

Shaked, M., and J. G. Shanthikumar, 2007. *Stochastic Orders.* New York: Springer.

Swiss Federal Office of Private Insurance. 2006. *Technical Document on the Swiss Solvency Test*. Berne: FINMA.

Tan, K. S., C. Weng, and Y. Zhang, 2009. VaR and CTE Criteria for Optimal Quota-Share and Stop-Loss Reinsurance. *North American Actuarial Journal* 13(4): 459–482.

Van Heerwaarden, A. E., R. Kaas, and M. J. Goovaerts, 1989. Optimal Reinsurance in Relation to Ordering of Risks. *Insurance: Mathematics and Economics* 8(1): 11–17.

Venter, G. G. 1991. Premium Calculation Implications of Reinsurance without Arbitrage. *Astin Bulletin* 21(2): 223–230.

Wang, S. 2002. A Set of New Methods and Tools for Enterprise Risk Capital Management and Portfolio Optimization. *CAS Summer Forum, Dynamic Financial Analysis Discussion Papers*, 43–78.

Wright, R. 1987. Expectation Dependence of Random Variables, with an Application in Portfolio Theory. *Theory and Decision* 22(2): 111–124.

Young, V. R. 1999. Optimal Insurance under Wang's Premium Principle. *Insurance: Mathematics and Economics* 25(2): 109–122.

Young, V. R. 2004. Premium Principles. In *Encyclopedia of Actuarial Science*, vol 3, ed. J. Teugels and B. Sundt. New York: John Wiley & Sons.

Zhu, W., K. S. Tan, and L. Porth 2016. *Agricultural Reinsurance Ratemaking: Development of a New Premium Principle*. Working paper. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2835481

Zhuang, S. C., C. Weng, K. S. Tan, and H. Assa, 2016. Marginal Indemnification Function Formulation for Optimal Reinsurance. *Insurance: Mathematics and Economics* 67: 65–76.

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APPENDIX. PROOFS

This appendix collects the proofs to the theorems and propositions discussed in the article.

A.1. Proof of Theorem 3.1

For an admissible ceded loss function $f(x) \in \mathfrak{C}$ and by the definition of $\mathcal{L}_f^{VaR_\alpha}(X)$ as given in (2.6), we have

$$\begin{aligned}
&\mathcal{L}_f^{VaR_\alpha}(X) \\
&\quad = (1-\delta)\mathbb{E}[R_f(X)] + \delta VaR_\alpha(R_f(X)) + \mathbb{E}[f(X) \times \mathcal{M}] \\
&\quad = (1-\delta)\mathbb{E}[X] + \delta VaR_\alpha(X) + \mathbb{E}[f(X) \times \mathcal{M}] - (1-\delta)\mathbb{E}[f(X)] - \delta f(VaR_\alpha(X)).
\end{aligned} \tag{A.1}$$

The first equality follows from the liability decomposition (2.5), the definition of the economic premium principle (2.3), and the translation invariant property of VaR. The second equality follows from (2.10) and $X = f(X) + R_f(X)$.

Similar to (3.2), it is easy to show that

$$\mathbb{E}[f(X)] = \int_0^\infty S_X(t)\, df(t) \tag{A.2}$$

and

$$f\left(VaR_\alpha(X)\right) = \int_0^\infty \mathbb{I}_{\{VaR_\alpha(X)>t\}}\, df(t) = \int_0^\infty \mathbb{I}_{\{S_X(t)>\alpha\}}\, df(t), \tag{A.3}$$

where the last equality can be justified using (2.9).

Consequently, (A.1) simplifies to

$$\mathcal{L}_f^{VaR_\alpha}(X) = (1-\delta)\mathbb{E}[X] + \delta VaR_\alpha(X) + \int_0^\infty \phi(t)\, df(t),$$

where $\phi(t)$ corresponds to (3.4). Note that $f'(t) \in [0, 1]$ for any $t \geq 0$. Thus, the admissible ceded loss function that minimizes $\mathcal{L}_f^{VaR_\alpha}(X)$ is given by (3.3), and this completes the proof.

A.2. Proof of Theorem 3.2

Using the same arguments for establishing (A.1), we have, for an admissible ceded loss function $f(x) \in \mathfrak{C}$,

$$\begin{aligned}
&\mathcal{L}_f^{CVaR_\alpha}(X) \\
&\quad = (1-\delta)\mathbb{E}[R_f(X)] + \delta \times CVaR_\alpha(R_f(X)) + \Pi_{\mathcal{M}}(f(X)) \\
&\quad = (1-\delta)\mathbb{E}[X] + \delta CVaR_\alpha(X) + \mathbb{E}[\mathcal{M}f(X)] - (1-\delta)\mathbb{E}[f(X)] - \delta CVaR_\alpha(f(X)).
\end{aligned} \tag{A.4}$$

It follows from Lemma 2.1 in Zhuang et al. (2016) that

$$CVaR_\alpha(f(X)) = \int_0^\infty \min\{1, S_X(t)/\alpha\}\, df(t).$$

Substituting (3.2), (A.2) and the above equation into (A.4) leads to

$$\mathcal{L}_f^{CVaR_\alpha}(X) = (1-\delta)\mathbb{E}[X] + \delta CVaR_\alpha(X) + \int_0^\infty \psi(t)\, df(t),$$

where $\psi(t)$ is defined in (3.6). As a consequence, $f_{cv}^*(x)$ given in (3.5) is a solution to the optimal reinsurance problem (2.12), and the proof is complete.

A.3. Proof of Proposition 4.1

If $\mathcal{M}$ is NED with respect to X, it follows from Definition 4.1 that

$$S_X^{\mathcal{M}}(t) = 1 - \mathbb{E}[\mathcal{M}\mathbb{I}_{\{X\le t\}}] = 1 - \mathbb{P}(X \le t) \times \mathbb{E}\left[\mathcal{M}|X \le t\right] \le S_X(t), \tag{A.5}$$

which in turn implies

$$\psi(t) \le -\delta \min\left\{F_X(t), S_X(t)(1/\alpha - 1)\right\} \le 0, \quad \forall t \ge 0.$$

Consequently, using Theorem 3.2, we get that the optimal ceded loss function under CVaR risk measure is given by $f_{cv}^*(x) = x$, as required.

A.4. Proof of Proposition 4.2

Note that similarly to establishing (A.5), we have $S_X^{\mathcal{M}}(t) \ge S_X(t)$ for any $t \ge 0$ if $\mathcal{M}$ is PED with respect to X. Moreover, it follows from (2.9) that $\mathbb{I}_{\{S_X(t)>\alpha\}} = 0$ for any $t \ge VaR_\alpha(X)$. Consequently, using (3.4), we have

$$\phi(t) = S_X^{\mathcal{M}}(t) - (1-\delta)S_X(t) \ge 0, \quad \forall t \ge VaR_\alpha(X),$$

which in turn implies $f_v^*(x) = f_v^*(VaR_\alpha(X))$ for all $x \ge VaR_\alpha(X)$.

On the other hand, if $\mathcal{M}$ is NED with respect to X, both (2.9) and (A.5) can be used again to show that

$$\phi(t) = S_X^{\mathcal{M}}(t) - (1-\delta)S_X(t) - \delta \le -\delta F_X(t) \le 0, \quad \forall t < VaR_\alpha(X).$$

Consequently, it follows from Theorem 3.1 that $f_v^*(x) = x$ for any $0 \le x \le VaR_\alpha(X)$. The proof is thus complete.

A.5. Proof of Proposition 4.3

By simple algebra, we can rewrite $S_X^{\mathcal{M}}(t)$ as

$$\begin{aligned} S_X^{\mathcal{M}}(t) &= \mathbb{E}\left[\int_0^\infty \mathbb{I}_{\{\mathcal{M}>y\}}\, dy \times \mathbb{I}_{\{X>t\}}\right] \\ &= \int_0^\infty \mathbb{P}(\mathcal{M} > y, X > t)\, dy \\ &= \int_0^\infty S_X(t) - F_{\mathcal{M}}(y) + C_{\mathcal{M},X}(F_{\mathcal{M}}(y), F_X(t))\, dy, \end{aligned} \tag{A.6}$$

where the second equality is due to Fubini's Theorem. Thus, if $C_{\mathcal{M}_1,X}(u,v) \ge C_{\mathcal{M}_2,X}(u,v)$ for any $(u,v) \in [0,1]^2$, we must have

$$S_X^{\mathcal{M}_1}(t) \ge S_X^{\mathcal{M}_2}(t), \quad \forall t \ge 0. \tag{A.7}$$

Thus the required result can be easily obtained from (3.3) and (3.5), and the proof is complete.

A.6. Proof of Proposition 4.4

Since it is assumed that $(\mathcal{M}_i, X)$ for $i = 1, 2$ have the same copula $C(u,v)$, it follows from Sklar's Theorem that there must exist uniformly distributed random variables U and V such that $(\mathcal{M}_i, X)$ has the same distribution with $(F_{\mathcal{M}_i}^{-1}(U), F_X^{-1}(V))$, where $F_Y^{-1}(t)$ is a generalized inverse function of $F_Y(t)$. Thus, $S_X^{\mathcal{M}_i}(t)$ can be rewritten as

$$S_X^{\mathcal{M}_i}(t) = \mathbb{E}\left[F_{\mathcal{M}_i}^{-1}(U)\mathbb{I}_{\{F_X^{-1}(V)>t\}}\right] = \mathbb{E}\left[F_{\mathcal{M}_i}^{-1}(U)\kappa_t(U)\right], \tag{A.8}$$

where

$$\kappa_t(u) \triangleq \mathbb{P}\left(F_X^{-1}(V) > t | U = u\right) \quad \text{for all } u \in [0,1], \quad t \ge 0.$$

If $\frac{\partial C(u,v)}{\partial u}$ is nonincreasing in u for every $v \in [0,1]$, and noting that $F_X^{-1}(y)$ is a nondecreasing and right-continuous function, we have $\kappa_t(u)$ is nondecreasing in u. Further, it is easy to see that $F_{\mathcal{M}_i}^{-1}(U)$ and $\kappa_t(U)$ are comonotonic. Since it is assumed that $\mathcal{M}_1 \leq_{cx} \mathcal{M}_2$, then Theorem 4.5 in Müller and Scarsini (2001) implies

$$\left(F_{\mathcal{M}_1}^{-1}(U), \kappa_t(U)\right) \leq_{dcx} \left(F_{\mathcal{M}_2}^{-1}(U), \kappa_t(U)\right),$$

where $(X_1, Y_1) \leq_{dcx} (X_2, Y_2)$ is defined by

$$\mathbb{E}[h(X_1, Y_1)] \leq \mathbb{E}[h(X_2, Y_2)]$$

for any directionally convex function $h(x, y)$, provided that the expectations exist.[2] Fortunately, it is easy to verify that $h(x, y) = xy$ is directionally convex, so that we have

$$\mathbb{E}\left[F_{\mathcal{M}_1}^{-1}(U)\kappa_t(U)\right] \leq \mathbb{E}\left[F_{\mathcal{M}_2}^{-1}(U)\kappa_t(U)\right],$$

which leads to the final result by using (A.8).

Now consider $\frac{\partial C(u,v)}{\partial u}$ is nondecreasing in u for every $v \in [0,1]$ so that $\kappa_t(u)$ is a nonincreasing function for any $t \geq 0$. Using similar arguments, we obtain $\mathbb{E}[F_{\mathcal{M}_1}^{-1}(U)(1-\kappa_t(U))] \leq \mathbb{E}[F_{\mathcal{M}_2}^{-1}(U)(1-\kappa_t(U))]$ for each $t \geq 0$. Further, using (A.8) and $\mathbb{E}[\mathcal{M}_i] = 1$, we have

$$S_X^{\mathcal{M}_i}(t) = 1 - \mathbb{E}\left[F_{\mathcal{M}_i}^{-1}(U)\,(1-\kappa_t(U))\right],$$

which in turn implies $S_X^{\mathcal{M}_1}(t) \geq S_X^{\mathcal{M}_2}(t)$. As a result, the insurer would cede less risk under the market factor $\mathcal{M}_1$ than under $\mathcal{M}_2$. The proof is thus complete.

A.7. Proof of Corollary 4.1

If Z is a constant, the market factor is always equal to one for any parameter, and hence the result is trivial. Thus, we assume $\mathbb{P}(Z = \mathbb{E}[Z]) < 1$ in the remaining part of the proof.

We first verify the result for Esscher premium principle. Specifically, we define

$$h_t(z) \triangleq e^{tz}/\mathbb{E}[e^{tZ}], \quad \forall z \geq 0,$$

which is obviously a strictly increasing function for any $t > 0$. For any $0 \leq t_2 < t_1$ with $\mathbb{E}[e^{t_1 Z}] < \infty$, let

$$A(z) \triangleq h_{t_1}(z)/h_{t_2}(z) = e^{(t_1-t_2)z} \times \frac{\mathbb{E}[e^{t_2 Z}]}{\mathbb{E}[e^{t_1 Z}]},$$

which is a strictly increasing continuous function with $A(0) < 1$ and $\lim_{z\to\infty} A(z) = \infty$. Thus, there must exist a $z_0 > 0$ such that $h_{t_1}(z) \leq h_{t_2}(z)$ for any $0 \leq z \leq z_0$ and $h_{t_1}(z) \geq h_{t_2}(z)$ for each $z > z_0$. In other words, the function $h_{t_1}(z)$ up-crosses the function $h_{t_2}(z)$. Noting that $\mathbb{E}[h_t(Z)] = 1$, we can get $h_{t_2}(Z) \leq_{cx} h_{t_1}(Z)$ by using Lemma 3 in Ohlin (1969). If $X \uparrow_{SI} Z$, then $B_x(z) \triangleq \mathbb{P}(X > x|Z = z)$ is nondecreasing over the support of Z for any $x \geq 0$. Using the same arguments as that in the proof of Proposition 4.4, we have

$$\mathbb{E}\left[h_{t_1}(Z)\mathbb{I}_{\{X>x\}}\right] = \mathbb{E}\left[h_{t_1}(Z)B_x(Z)\right] \geq \mathbb{E}\left[h_{t_2}(Z)B_x(Z)\right] = \mathbb{E}\left[h_{t_2}(Z)\mathbb{I}_{\{X>x\}}\right],$$

[2] A two-dimensional function $h(x, y)$ is called to be directionally convex if it satisfies

$$h(x_1, y_1) + h(x_2, y_2) \leq h(\min\{x_1, x_2\}, \min\{y_1, y_2\}) + h(\max\{x_1, x_2\}, \max\{y_1, y_2\})$$

and is convex in each coordinate when the other coordinate is fixed.

which together with Theorems 3.1 and 3.2 leads to that the insurer would cede less risk for a larger t. Otherwise, if $X \uparrow_{SI} -Z$, then $B_x(z)$ is nonincreasing over the support of Z. Similarly, we have

$$\mathbb{E}\left[h_{t_1}(Z)\mathbb{I}_{\{X>x\}}\right] = 1 - \mathbb{E}\left[h_{t_1}(Z)\left(1 - B_x(Z)\right)\right] \le 1 - \mathbb{E}[h_{t_2}(Z)(1 - B_x(Z))] = \mathbb{E}\left[h_{t_2}(Z)\mathbb{I}_{\{X>x\}}\right],$$

which implies that optimal ceded loss functions are larger for a higher t.

The proofs for size-biased, Aumann-Shapley, and proportional hazard premium principles are similar and hence are omitted. We now proceed to consider the excess-of-loss premium principle. It is easy to see that the function $\mathbb{I}_{\{z\ge t_1\}}/\mathbb{P}(Z \ge t_1)$ up-crosses the function $\mathbb{I}_{\{z\ge t_2\}}/\mathbb{P}(Z \ge t_2)$ for any $0 \le t_2 < t_1$ with $\mathbb{P}(Z \ge t_1) > 0$. Thus, it follows from Lemma 3 in Ohlin (1969) that

$$\mathbb{I}_{\{Z\ge t_2\}}/\mathbb{P}(Z \ge t_2) \le_{cx} \mathbb{I}_{\{Z\ge t_1\}}/\mathbb{P}(Z \ge t_1).$$

Noting that $\mathbb{I}_{\{z\ge t\}}/\mathbb{P}(Z \ge t)$ is nondecreasing in z for each $t \ge 0$, we obtain the result following the arguments similar to the Esscher premium principle. Furthermore, from the definition of VaR, we can see that $VaR_p(Z)$ is nonincreasing in p. Hence the analysis for the excess-of-loss premium principle can similarly be applied to CTE and modified tail covariance premium principles, so their proofs are not presented.

Finally, we show the result for Kamps premium principle. In this case it is necessary only to prove $K_{t_1}(Z) \le_{cx} K_{t_2}(Z)$ for any $0 \le t_2 < t_1$, where

$$K_t(z) \triangleq (1 - e^{-tz})/\mathbb{E}[1 - e^{-tZ}], \quad \forall z \ge 0.$$

Obviously, $K_t(z)$ is a nondecreasing continuous function for any $t \ge 0$, and

$$\gamma(z) \triangleq K_{t_1}(z)/K_{t_2}(z) = \frac{\int_0^z e^{-t_1 y}\mathrm{d}y}{\int_0^z e^{-t_2 y}\,dy} \times \frac{\mathbb{E}\left[\int_0^Z e^{-t_2 y}\mathrm{d}y\right]}{\mathbb{E}\left[\int_0^Z e^{-t_1 y}\,dy\right]}.$$

Thus, we have $\gamma(0) = \frac{\mathbb{E}[\int_0^Z e^{-t_2 y}\,dy]}{\mathbb{E}[\int_0^Z e^{-t_1 y}\,dy]} > 1$,

$$\gamma'(z) = \frac{\gamma(z)e^{-t_2 z}}{\int_0^z e^{-t_1 y}\,dy} \times \left(\frac{e^{-t_1 z}}{e^{-t_2 z}} - \frac{\int_0^z e^{-t_1 y}\,dy}{\int_0^z e^{-t_2 y}\,dy}\right) < 0,$$

and hence $K_{t_2}(z)$ up-crosses $K_{t_1}(z)$. Noting that $\mathbb{E}[K_t(Z)] = 1$, it follows from Lemma 3 in Ohlin (1969) that $K_{t_1}(Z) \le_{cx} K_{t_2}(Z)$. The proof is now complete.

A.8. Proof of Proposition 5.1

If the market factor $\mathcal{M}$ and the loss X have a bivariate copula of the form (5.1), (A.6) together with $\mathbb{E}[\mathcal{M}] = 1$ leads to

$$S_X^{\mathcal{M}}(t) = S_X(t)\left(1 + F_X(t)Q_\lambda^{\mathcal{M}}\right), \tag{A.9}$$

where the constant $Q_\lambda^{\mathcal{M}}$, which is defined in (5.3), is subject to

$$\left|Q_\lambda^{\mathcal{M}}\right| \le \int_0^\infty \left|\lambda\left(F_{\mathcal{M}}(y)\right) - \lambda(1)\right| dy \le \int_0^\infty S_{\mathcal{M}}(y)\,dy = 1.$$

The above inequality follows from the Lipschitz condition (5.2) and $\lambda(1) = 0$. As a consequence, we have

$$\mathbb{E}[\mathcal{M}|X \le t] = F_X^{\mathcal{M}}(t)/\mathbb{P}(X \le t) = 1 - S_X(t)Q_\lambda^{\mathcal{M}}.$$

The result is easily obtained by using Definition 4.1.

A.9. Proof of Proposition 5.2

If $\mathcal{M}$ and X have a copula (5.1), it follows from (3.4) and (A.9) that

$$\phi(t) = \begin{cases} F_X(t)\left(Q_\lambda^{\mathcal{M}} S_X(t) - \delta\right), & 0 \le t < VaR_\alpha(X); \\ S_X(t)\left(Q_\lambda^{\mathcal{M}} F_X(t) + \delta\right), & t \ge VaR_\alpha(X). \end{cases}$$

If $Q_\lambda^{\mathcal{M}} \in [-1, -\delta)$, we can see from the above equation that $\phi(t) \le 0$ for $0 \le t < VaR_\alpha(X)$. Moreover, it follows from (2.9) that $Q_\lambda^{\mathcal{M}} F_X(t) + \delta \le 0$ for $t \ge VaR_\alpha(X)$ is equivalent to $t \ge VaR_{\min\{\alpha, 1+\frac{\delta}{Q_\lambda^{\mathcal{M}}}\}}(X)$. Thus, the result follows by using (3.3). Else if $Q_\lambda^{\mathcal{M}} \in [-\delta, \delta]$, it is easy to see that $\phi(t)$ is negative for $0 \le t < VaR_\alpha(X)$ and is non-negative for $t \ge VaR_\alpha(X)$. Thus, it follows from (3.3) that $f_v^*(x) = \min\{x, VaR_\alpha(X)\}$. Otherwise, if $Q_\lambda^{\mathcal{M}} \in (\delta, 1]$, it is trivial that $\phi(t) \ge 0$ for each $t \ge VaR_\alpha(X)$. Moreover, using (2.9) and the assumption of $\delta \ge \alpha$, we obtain that $Q_\lambda^{\mathcal{M}} S_X(t) - \delta \le 0$ for $0 \le t \le VaR_\alpha(X)$ is equivalent to $t \in [VaR_{\frac{\delta}{Q_\lambda^{\mathcal{M}}}}(X), VaR_\alpha(X)]$. Thus, $f_v^*(x)$ is a one-layer reinsurance.

Similarly, (3.6) and (A.9) lead to

$$\psi(t) = \begin{cases} F_X(t)\left(Q_\lambda^{\mathcal{M}} S_X(t) - \delta\right), & 0 \le t < VaR_\alpha(X); \\ S_X(t)\left(Q_\lambda^{\mathcal{M}} F_X(t) - \delta\left(\frac{1}{\alpha} - 1\right)\right), & t \ge VaR_\alpha(X). \end{cases}$$

If $Q_\lambda^{\mathcal{M}} \in [-1, 0]$, it is obvious that $\psi(t) \le 0$ for $t \ge 0$ such that $f_{cv}^*(x) = x$. Otherwise, if $Q_\lambda^{\mathcal{M}} > 0$, we can see from the above analysis that $Q_\lambda^{\mathcal{M}} S_X(t) - \delta \le 0$ for $0 \le t < VaR_\alpha(X)$ if and only if $t \in [VaR_{\frac{\delta}{Q_\lambda^{\mathcal{M}}}}(X), VaR_\alpha(X))$. Moreover, similarly using (2.9), we obtain $Q_\lambda^{\mathcal{M}} F_X(t) - \delta(\frac{1}{\alpha} - 1) < 0$ is equivalent to $t < VaR_{1-\frac{\delta}{Q_\lambda^{\mathcal{M}}}(\frac{1}{\alpha}-1)}(X)$. Noting that $\delta \ge \alpha$, we have $1 - \frac{\delta}{Q_\lambda^{\mathcal{M}}}(\frac{1}{\alpha} - 1) \le \alpha$. Consequently, the result follows from (3.5) for this case, and the proof is thus complete.

A.10. Proof of Equation (5.4)

Recall that under the FGM copula $\lambda(u) = \theta u(1-u)$ for a $\theta \in [-1, 1]$. Thus, we have from (5.3)

$$\begin{aligned} Q_\lambda^{\mathcal{M}} &= \theta \int_0^\infty F_{\mathcal{M}}(y)\left(1 - F_{\mathcal{M}}(y)\right) dy = \theta\left(\mathbb{E}[\mathcal{M}] - \int_0^\infty \left(1 - F_{\mathcal{M}}(y)\right)^2 dy\right) \\ &= \theta\left(1 - \mathbb{E}\left[\min\{\mathcal{M}, \tilde{\mathcal{M}}\}\right]\right) = \theta\left(1 - \mathbb{E}\left[e^{\eta \times \min\{N, \tilde{N}\} - \frac{\eta^2}{2}}\right]\right), \end{aligned}$$

where $(\tilde{\mathcal{M}}, \tilde{N})$ is an independent copy of $(\mathcal{M}, N)$. To complete the proof, note that the probability density function of the random variable $\min\{N, \tilde{N}\}$ is given by

$$\frac{2}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} \mathcal{N}(-t), \quad t \in \mathbb{R}.$$

so that

$$\begin{aligned} Q_\lambda^{\mathcal{M}} &= \theta\left(1 - 2\int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{(t-\eta)^2}{2}} \mathcal{N}(-t)\, dt\right) \\ &= \theta\left(1 - 2\mathbb{P}(N > \tilde{N} + \eta)\right) = \theta\left(1 - 2\mathbb{P}\left(\frac{N - \tilde{N}}{\sqrt{2}} > \eta/\sqrt{2}\right)\right) \\ &= \theta\left(1 - 2\mathcal{N}\left(-\eta/\sqrt{2}\right)\right). \end{aligned}$$

The last equality follows trivially from the fact that $\frac{N-\tilde{N}}{\sqrt{2}}$ is a standard normal random variable, and this completes the proof.